



Darren K. Woodland, Jr.

Experimental Media Artist & Designer-Technologist

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PROFESSIONAL SUMMARY

PhD researcher, media technologist, and practicing artist with 8 years of experience building real-time interactive systems, XR environments, and live audiovisual performance work. Technical fluency in Unity, Unreal Engine, physical computing, computer vision, and creative coding in C#, Python, and C++ is grounded in a sustained studio practice that includes over a decade as a performing musician, drawing and illustration, and hands-on fabrication. Work spans interactive installation, game development, staged performance, and academic research, with exhibitions and presentations at Ars Electronica, SXSW, and SIGGRAPH. Active as both researcher and instructor at the graduate and undergraduate levels.

EDUCATION

[Drexel University](#), Philadelphia, PA

Ph.D. in Digital Media

January 2022 – Present

Research: Afrofuturism, real-time audio, immersive experience design, gesture data, motion-to-sound mappings, audiovisual sonification, narrative co-creation

Graduate Minor in Computer Science

September 2022 – January 2025

Focus: Software design, computer vision, machine learning, audio programming

[North Carolina State University](#), Raleigh, NC

M.A.D. in Experimental Media Arts

August 2018 – December 2020

Research: Sound, interaction design, extended realities

Graduate Certificate in Digital Humanities

May 2019 – December 2020

Focus: Games in social contexts, ludic networks, game design

[University of South Carolina](#), Columbia, SC

B.A. in Media Arts & Art Studio

August 2013 – December 2017

Media Arts: Animation, narrative writing, game design

Art Studio: Graphic design, illustration, drawing

WORK EXPERIENCE

[Robotics, Automation, and Dance \(RAD\) Lab](#), Philadelphia, PA

Artist in Residence

May 2025 – Present

- Design and prototype gesture-based interaction systems for human-robot interaction (HRI) research, focusing on symbolic movement encoding, embodied interaction, and gesture-driven data modalities
- Produce grant documentation and interactive reference materials for a funded project

Drexel University, Philadelphia, PA

PhD Researcher & Developer

January 2022 – Present

- Lead technical architecture and cross-functional development for *BioreactAR*, an interactive AR learning simulation built in Unity 6 with Quest 3 passthrough
- Served as lead technical collaborator on *Mnemosi* (WOMB: The Black Wealth Project, 2024) and designed motion-derived animation systems for *Gr[i]ef* (Mandell Theater, 2024), building real-time interactive projection and performance systems for live staged work
- Developed archival heritage preservation workflows, including hardware-to-software pipelines for Gaussian Splat capture and immersive exhibition of digital museum artifacts

Adjunct Instructor & Graduate Teaching Fellow

January 2022 – Present

- Design and teach 5 undergraduate and cross-listed graduate courses in scripting, games and development, and technical art: DIGM 131, GMAP 231, GMAP 260, GMAP 345/CS 345/DIGM 545, and GMAP 395

Version Control (Perforce) Systems Administrator

January 2024 – Present

- Maintain Perforce server infrastructure and repository configuration for faculty and students college-wide
- Author technical documentation and support benchmarks to streamline multi-user production pipelines across Digital Media departments

Colorado Conservatory of Dance, Denver, CO

Freelance Multimedia Artist

January 2024 – July 2024

- Designed audio systems, animations, and real-time Unreal Engine visuals for a live stage dance production
- Integrated and edited all multimedia content in DaVinci Resolve for live stage delivery

UNCSA METL, Winston-Salem, NC

Art Director & Technical Artist

April 2021 – October 2021

- Directed art direction and asset development for an Unreal Engine VR project, producing 200+ assets and integrating spatial audio and sound design into the production pipeline

The IntelliMedia Group (CEI), Raleigh, NC

Digital Artist & Graduate Research Assistant

May 2019 – December 2020

- Designed, modeled, and animated 2D and 3D character models, props, and environmental assets for educational games and VR training applications

NCSU College of Design, Raleigh, NC

Graduate Teaching Assistant

August 2018 – May 2019

- Instructed industry-standard design software and workflows for undergraduate students

PUBLICATIONS & EXHIBITIONS

Digital Twin (Art Exhibition). *Digital Twin: Transforming Human Experience Through Digital Rendering*. 2025.

SIGGRAPH Asia (Paper). *Plumes and Pixels: Bridging Digital and Traditional Art in SciArt Education*. 2024.

Media Arts and Design Conference (Paper & Presentation). *Motifs of the Future Metamorphosis: Afrofuturism in Media Arts and Design*. 2024.

Mandell Theater (Dance Performance). *Gr[i]ef*. 2024.

Colorado Conservatory of Dance (Dance Performance). *Home is in Me*. 2024.

Ars Electronica (Presentation). *Echoes of Harmonic Threads*. 2023.

SXSW, SIGGRAPH, Bucheon Film Festival, Pico VR. *Black Ice VR*. 2022.

Audiokinetic Blog. *Cybernetic Thrills: Black Ice VR*. 2022.

NCSU M.A.D. Thesis. *On Soundscape: Listening as Interaction*. 2020.

AIGA of SC InShow 22 Awards. *Pop! Art Book Covers*. 2017.

Knowitall.org (Animated Video). *Scientific SC*. 2017.

SKILLS

Programming

Visual programming, C#, C, C++, Java, Python, JavaScript, Lua

Game Development

Unity, Unreal Engine, real-time graphics optimization

3D and Simulation

Houdini, Maya, Reallusion

Interactive Media

TouchDesigner, Isadora, PureData, vvvv

Audio Development

Reaper, Max/MSP, PlugData, MetaSounds, Wwise

XR Platforms

VR, AR, Mixed Reality

Computer Vision and Tracking

Nuitrack, MediaPipe, OpenCV, mocap pipelines, pose estimation

Hardware and Physical Computing

ESP32, Teensy, Arduino, Kinect v2, piezoelectric sensors, microphones, hydrophones

Version Control

Perforce, Git, Plastic SCM (Unity), Jira

Media Production

DaVinci Resolve, After Effects, Premiere Pro, Adobe Creative Cloud, Substance

Design Principles

Color theory, graphic design, visual storytelling, design thinking, iterative design

Core Competencies

Cross-disciplinary collaboration, technical documentation, project management, research methodology, performance direction